Technical Document #504: Retrieval Augmented Generation - Comprehensive Overvi

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Vector databases store dense embeddings for semantic search.
- Query decomposition breaks complex queries into simpler sub-queries.